

HORNSBY GIRLS HIGH SCHOOL



Mathematics

Year 12 Higher School Certificate
Trial Examination Term 3 2013

STUDENT NUMBER: _____

General Instructions

- Reading Time – 5 minutes
- Working Time – 3 hours
- Write using black or blue pen
Black pen is preferred
- Board-approved calculators may be used
- A table of standard integrals is provided separately
- In Questions 11 – 16, show relevant mathematical reasoning and/or calculations
- Marks may be deducted for untidy and poorly arranged work
- Do not use correction fluid or tape
- Do not remove this paper from the examination

Total marks – 100

Section I Pages 3 – 5

10 marks

Attempt Questions 1 – 10

Answer on the Objective Response Answer Sheet provided

Section II Pages 6 – 15

90 marks

Attempt Questions 11 – 16. Start each question in a new writing booklet. Write your student number on every writing booklet

Question	1-10	11	12	13	14	15	16	Total
Total	/10	/15	/15	/15	/15	/15	/15	/100

This assessment task constitutes 45% of the Higher School Certificate Course School Assessment

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Section I

10 marks

Attempt Questions 1 – 10

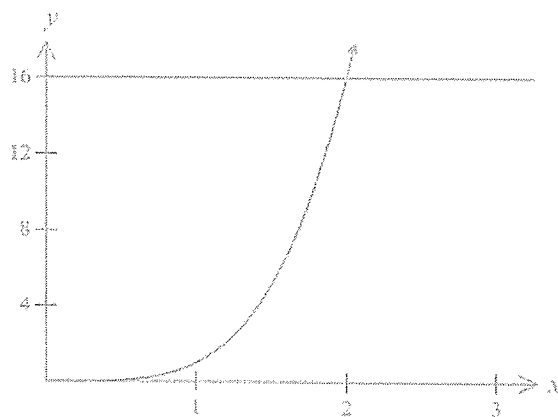
Allow about 15 minutes for this section

Use the Objective Response answer sheet for Questions 1 – 10

- 1 What is 25.09582 correct to 4 significant figures?
- (A) 25.09
(B) 25.10
(C) 25.095
(D) 25.096
- 2 The solutions of $8x - x^2 < 0$ are:
- (A) $0 < x < 8$
(B) $x < 0, x < 8$
(C) $x < 0, x > 8$
(D) $x > 0, x < 8$
- 3 Factorise completely $a^3b - ab^3 - 4a^2 + 4b^2$
- (A) $ab(a^2 - b^2) - 4(a^2 + b^2)$
(B) $ab(a^2 - b^2) - 4(a^2 - b^2)$
(C) $(ab - 4)(a^2 - b^2)$
(D) $(a - b)(ab - 4)(a + b)$
- 4 If α and β are the roots of the equation $2x^2 + 5x - 4 = 0$, then $\alpha^2 + \beta^2 =$
- (A) $6\frac{1}{4}$
(B) $10\frac{1}{4}$
(C) $2\frac{1}{4}$
(D) 4

- 5 The period of the function $y = 3 \tan\left(\frac{x}{3}\right)$ is:
- (A) $\frac{\pi}{3}$
(B) π
(C) 3π
(D) 6π
- 6 What is the equation of the tangent to the curve $y = x^2 - 5x$ at the point $(1, -4)$?
- (A) $y = -3x - 1$
(B) $y = -3x - 7$
(C) $y = 3x + 7$
(D) $y = 3x - 7$
- 7 Solve $\sin 2\theta = \frac{-\sqrt{3}}{2}$ in the domain $0^\circ \leq \theta \leq 180^\circ$:
- (A) $\theta = 60^\circ$
(B) No solutions
(C) $\theta = 120^\circ$
(D) $\theta = 120^\circ, 150^\circ$
- 8 What is the solution to the equation $\log_e(x+2) - \log_e x = \log_e 4$?
- (A) $\frac{2}{5}$
(B) $\frac{2}{3}$
(C) $\frac{3}{2}$
(D) $\frac{5}{2}$

- 9 A region in the diagram is bounded by the curve $y = x^4$, the y -axis and the line $y = 16$.



Which of the following expressions is correct for the volume of the solid of revolution when this region is rotated about the y -axis?

(A) $V = \pi \int_0^{16} y^{\frac{1}{2}} dy$

(B) $V = \pi \int_0^{16} x^8 dx$

(C) $V = \pi \int_0^2 y^{\frac{1}{2}} dy$

(D) $V = \pi \int_0^2 x^8 dx$

- 10 The value of $\sum_{n=2}^5 2n^2$ is:

(A) 50

(B) 108

(C) $205\frac{1}{32}$

(D) $362\frac{21}{32}$

End of Section I

Section II

90 marks

Attempt Questions 11 – 16

Allow about 2 hours and 45 minutes for this section

Answer each question in a new writing booklet. Extra writing booklets are available.

In Questions 11 – 16, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks) Start a new writing booklet

- (a) Evaluate $\sqrt[3]{\left(\frac{a}{b}\right)^b}$, correct to three decimal places, where $a = 3$ and $b = 2$. 2
- (b) Solve the equation $4x^2 = x$. 2
- (c) Solve the equation $|4 - x| = 2x$. 2
- (d) Express $\frac{1}{4 - \sqrt{13}}$ in the form $a + b\sqrt{13}$, where a and b are rational numbers. 2
- (e) Find $\int e^{3x} dx$. 1
- (f) Sketch the parabola $x^2 = -4y + 8$, showing the vertex, focus and directrix. 3
- (g) Differentiate x^3 from first principles. 3

Question 12 (15 marks) Start a new writing booklet

(a) Differentiate the following, with respect to x :

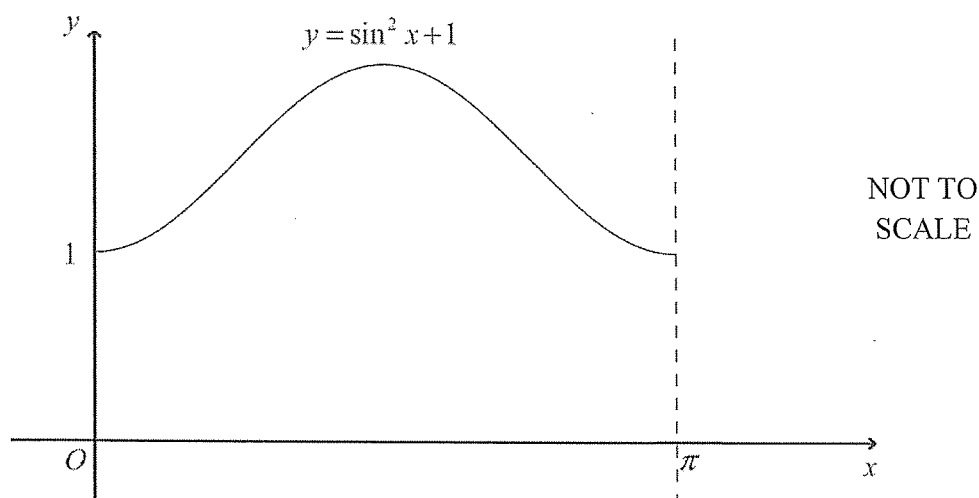
(i) $e^x \sin x$. 2

(ii) $\frac{\log_e x}{x^3}$. 2

(b) Find the exact value of $\int_0^{\frac{\pi}{4}} \frac{\cos x}{\sin x + 3} dx$. 3

(c) A stained-glass window can be modelled as the region bounded by the curve $y = \sin^2 x + 1$, the coordinate axes and $x = \pi$.

The graph of $y = \sin^2 x + 1$ for $0 \leq x \leq \pi$ is shown below.



(i) Copy and complete the table with exact values in your writing booklet. 1

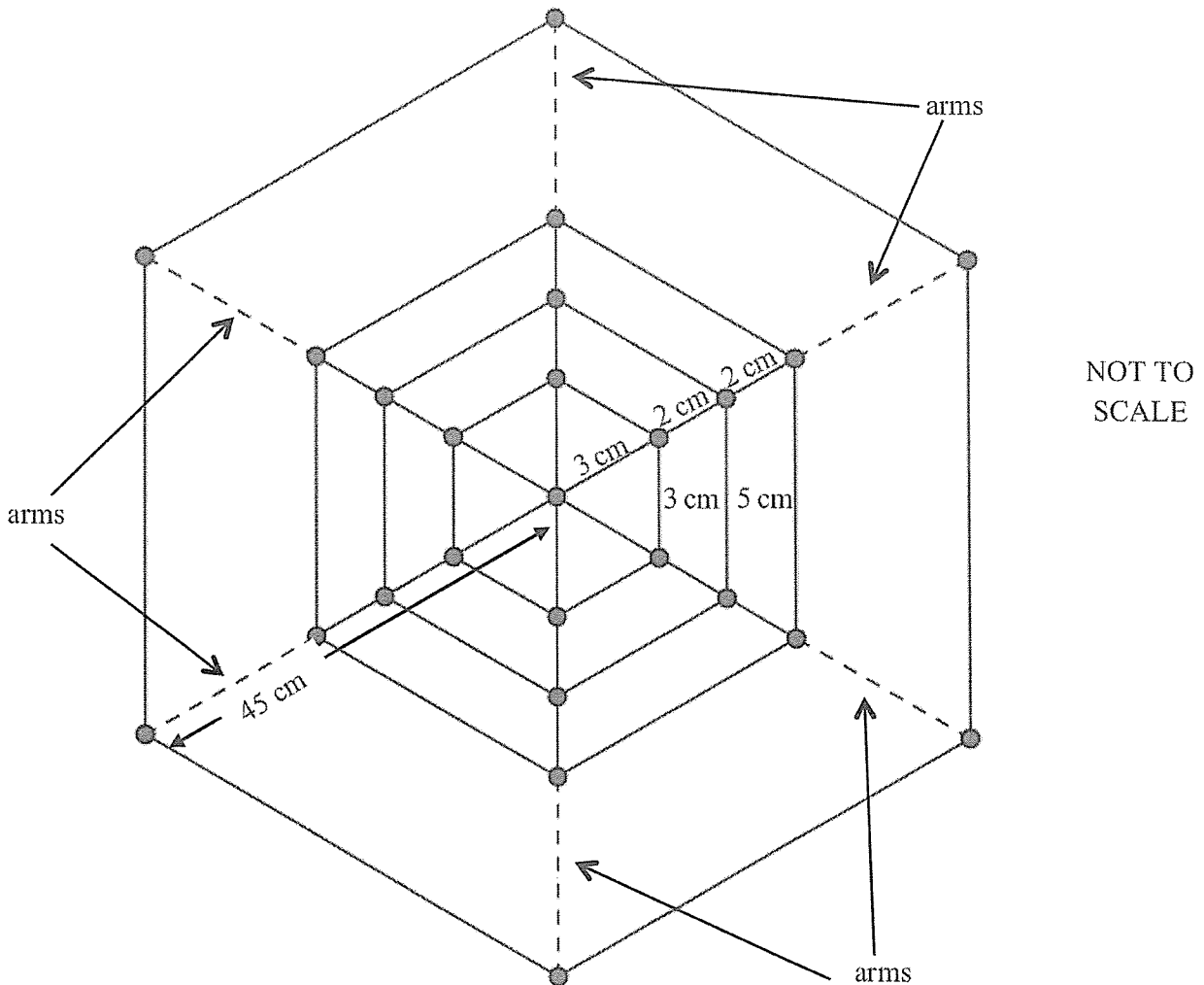
x	0	$\frac{\pi}{4}$	$\frac{\pi}{2}$	$\frac{3\pi}{4}$	π
y	1				1

(ii) Use Simpson's Rule with 5 function values to calculate the approximate area of glass needed for the window. Answer correct to 2 decimal places. 2

Question 12 continues on page 8

Question 12 (continued)

- (d) Incey Wincey spider makes a web in the shape of concentric regular hexagons. First he makes the 6 arms which are each 45 cm long. He then makes the sides of each regular hexagon. The first is 3 cm from the centre along each arm. Each successive regular hexagon is 2 cm further along the arm. The last hexagon is at the end of the arms.

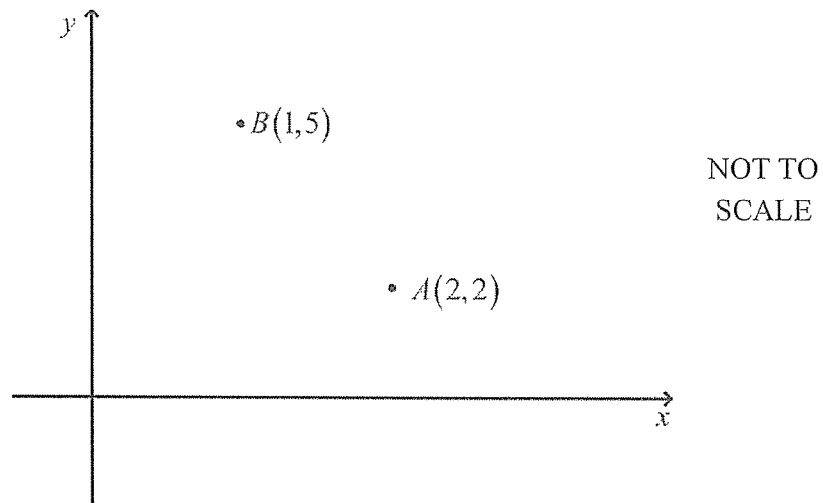


- | | | |
|------|--|---|
| (i) | How many regular hexagons does Incey Wincey create? | 2 |
| (ii) | Find the total length of the web, including the arms, created by Incey Wincey. | 3 |

End of Question 12

Question 13 (15 marks) Start a new writing booklet

- (a) The diagram below shows two points $A(2,2)$ and $B(1,5)$ on the number plane.



Copy the diagram into your writing book.

- (i) Find the coordinates of M , the midpoint of AB . **1**
- (ii) Show that the equation of the perpendicular bisector of AB is $x - 3y + 9 = 0$. **2**
- (iii) Find the coordinates of the point C that lies on the y -axis and is equidistant from A and B . **1**
- (iv) The point D lies on the intersection of the line $y = 5$ and the perpendicular bisector $x - 3y + 9 = 0$. Find the coordinates of D , and mark the position of D on your diagram in your writing booklet. **1**
- (v) Find the area of triangle ABD . **2**

Question 13 continues on page 10

Question 13 (continued)

- (b) (i) Find the points of intersection of the parabola $y = x^2 + 3x - 5$ and the line $y = 2x + 1$. 2
- (ii) Hence find the area enclosed between the parabola $y = x^2 + 3x - 5$ and the line $y = 2x + 1$. 2
- (c) A school softball team has a probability of 0.2 of winning any match.
- (i) Find the probability the team wins exactly one of its first two matches. 2
- (ii) What is the least number of consecutive matches the team must play to be 90% certain that it will win **at least one** match? 2

End of Question 13

Question 14 (15 marks) Start a new writing booklet

(a) Consider the curve $y = x^3 + 4x^2 - 3x + 2$.

(i) Find $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$. 2

(ii) Find the coordinates of the stationary points and determine their nature. 2

(iii) Find the coordinates of any points of inflexion. 2

(iv) Sketch the graph of $y = x^3 + 4x^2 - 3x + 2$, clearly showing any stationary points, points of inflexion and the y -intercept. 2

(v) Hence find the number of solutions to the equation $x^3 + 4x^2 - 3x + 2 = -1$. 1

(b) The mass M kg of a radioactive substance present after t years is given by $M = 20e^{-kt}$, where k is a positive constant. After 200 years, the mass has reduced to 10kg.

(i) What is the initial mass? 1

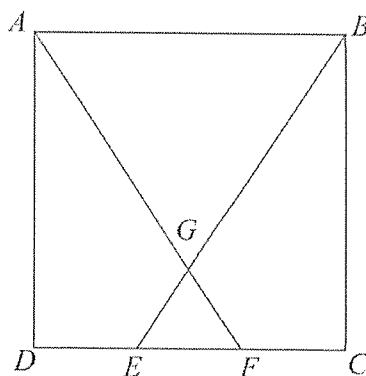
(ii) Find the exact value of k . 2

(iii) What amount of radioactive substance would remain after a period of 2000 years? 1

(c) For what values of k does the equation $x^2 - 2x + 3 = k$ have real roots? 2

Question 15 (15 marks) Start a new writing booklet

- (a) $ABCD$ is a square. Points E and F lie on DC such that $DE = CF$.



NOT TO
SCALE

Copy or trace the diagram into your writing booklet

- (i) Prove that $\triangle AFD \equiv \triangle BEC$. 3
- (ii) Hence or otherwise, prove that $GE = GF$. 1
- (b) The velocity of a particle is given by $\frac{dx}{dt} = t^2 - 5t + 6$, $t \geq 0$, where x is displacement in metres and t is time in seconds. Initially the particle is 3 metres to the left of the origin.
- (i) Find when the velocity of the particle is zero. 1
- (ii) Find the minimum velocity of the particle. 1
- (iii) Find the displacement x of the particle in terms of t . 2
- (iv) Find the distance travelled by the particle in the first 3 seconds. 2

Question 15 continues on page 13

Question 15 (continued)

- (c) Rita takes out a loan of \$30 000 to buy a new car. The loan is to be repaid over 5 years (60 months) in equal monthly repayments (Q) at the end of each month.

Reducible interest is charged at 9% per annum, calculated monthly.

Let $\$A_n$ be the amount owing after the n th payment.

- (i) Write an expression for the amount owing after 1 month, $\$A_1$. 1

- (ii) Show that $\$A_n = 30000(1.0075)^n - \frac{400Q(1.0075^n - 1)}{3}$. 2

- (iii) What will the monthly repayment be to the nearest dollar? 2

End of Question 15

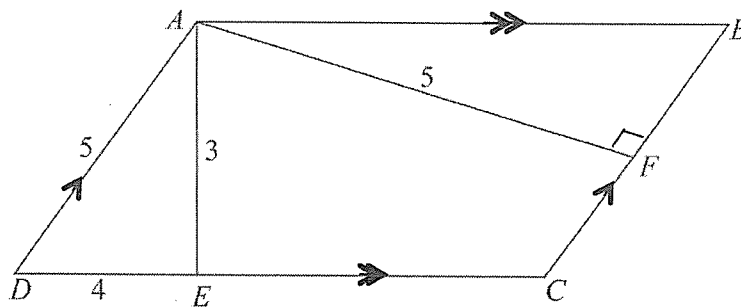
Question 16 (15 marks) Start a new writing booklet

(a) If $6x^2 - 11 \equiv A(x+2)^2 + Bx + C$, find the values of A , B and C . 2

(b) (i) Given that $a^2 + b^2 = 7ab$, show that $\left(\frac{a+b}{3}\right)^2 = ab$. 1

(ii) Hence, write $\log\left(\frac{a+b}{3}\right) - \frac{1}{4}(\log a + \log b)$ in simplest form. 2

- (c) In the diagram below, ABCD is a parallelogram. $AD = 5$, $AE = 3$, $DE = 4$, $AF = 5$ and $AF \perp BC$.



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SCALE

Copy or trace the diagram into to your writing booklet

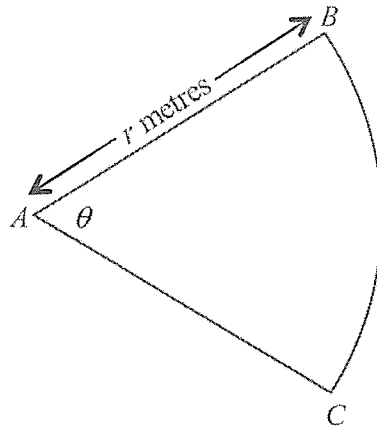
(i) Prove that $\angle ADE = \angle EAF$. 2

(ii) Find the exact length of EF . 2

Question 16 continues on page 15

Question 16 (continued)

- (d) In the figure below, AB and AC are radii of length r metres of a circle with centre A .
The arc BC of the circle subtends an angle of θ at A .
The perimeter of the figure ABC is 12 metres.



- (i) Show that the area Y square metres of the sector ABC is given by $Y = \frac{72\theta}{(\theta+2)^2}$. 2
- (ii) Hence, show that the maximum area of the sector is 9 square metres. 4

End of Paper

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Hornsby Girls High School Mathematics Trial Examination 2013 Solutions and Marking Criteria
Objective Response Questions 1 – 10

Solutions	Marking Criteria
Question 1 $25.0983 \approx 25.10 \text{ (4sf)}$ Option B	<i>Students should not be getting this wrong at this point! You must speak to your teacher if you need help with rounding.</i>
Question 2 $8x - x^2 < 0$ $x(8 - x) < 0$ $\therefore x < 0, x > 8$ Option C	
Question 3 $a^3b - ab^3 - 4a^2 + 4b^2 = ab(a^2 - b^2) - 4(a^2 - b^2)$ $= (a^2 - b^2)(ab - 4)$ $= (a - b)(a + b)(ab - 4)$ Option D	<i>A lot of students failed to recognise the difference of two squares.</i>
Question 4 $2x^2 + 5x - 4 = 0$ $\alpha + \beta = \frac{-5}{2} \quad \alpha\beta = -2$ $\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$ $= \frac{25}{4} + 4$ $= \frac{41}{4}$ $= 10\frac{1}{4}$ Option B	
Question 5 $T = \frac{\pi}{n}$ $= \pi \div \frac{1}{3}$ $= 3\pi$ Option C	<i>Many students just assumed the period was $\frac{2\pi}{n}$ for trig functions.</i>
Question 6 $y = x^2 - 5x$ $y' = 2x - 5$ When $x = 1$, $y' = -3$ Equation of tangent: $y + 4 = -3(x - 1)$ $y = -3x + 3 - 4$ $y = -3x - 1$ Option A	

Question 7 $\sin 2\theta = \frac{-\sqrt{3}}{2}$ $0^\circ \leq \theta \leq 180^\circ$ $0^\circ \leq 2\theta \leq 360^\circ \leftarrow$ $2\theta = 240^\circ, 300^\circ$ $\theta = 120^\circ, 150^\circ$ Option D	<i>Students should adjust the domain to show restrictions on 2θ.</i>
Question 8 $\ln(x+2) - \ln x = \ln 4$ $\ln\left(\frac{x+2}{x}\right) = \ln 4$ $\frac{x+2}{x} = 4$ $x+2 = 4x$ $2 = 3x$ $x = \frac{2}{3}$ Option B	
Question 9 $V = \pi \int_a^b x^2 dy$ $= \pi \int_0^{16} y^{\frac{1}{2}} dy$ Option A	
Question 10 $2(2^2 + 3^2 + 4^2 + 5^2) = 108$ Option B	

Question 11

Solutions	Marking Criteria
(a) $\sqrt[3]{\left(\frac{3}{2}\right)^2} = 1.31037\dots$ $= 1.310 \text{ (3dp)}$	
(b) $4x^2 = x$ $4x^2 - x = 0$ $x(4x - 1) = 0$ $x = 0, x = \frac{1}{4}$	
(c) $ 4 - x = 2x$ Case 1 Case 2: $4 - x = 2x$ $4 = 3x$ $x = \frac{4}{3}$ Testing Solutions When: $x = \frac{4}{3}$ $LHS = \left 4 - \frac{4}{3}\right $ $= \left \frac{8}{3}\right $ $= \frac{8}{3}$ $RHS = 2 \times \frac{4}{3}$ $= \frac{8}{3}$ $= RHS$ The only solution is $x = \frac{4}{3}$	<i>Too many students are not testing solutions.</i>
(d) $\frac{1}{4 - \sqrt{13}} \times \frac{4 + \sqrt{13}}{4 + \sqrt{13}} = \frac{4 + \sqrt{13}}{3}$ $= \frac{4}{3} + \frac{\sqrt{13}}{3}$ $a = \frac{4}{3}, b = \frac{1}{3}$	

(e)

$$\int e^{3x} dx = \frac{1}{3} \int 3e^{3x} dx$$
$$= \frac{1}{3} e^{3x} + C$$

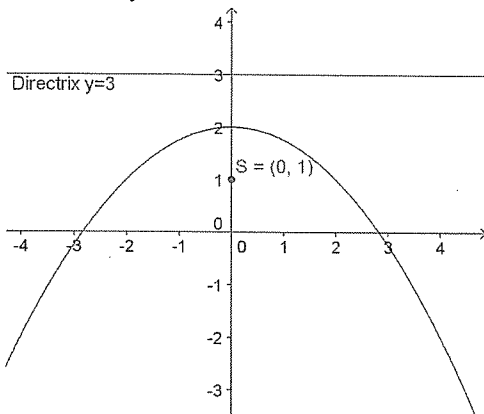
(f)

$$x^2 = -4y + 8$$

$$x^2 = -4(y - 2)$$

Focus: $S = (0, 1)$

Directrix $y = 3$



(g)

Change solutions

$$f(x) = x^3$$

$$f(x+h) = (x+h)^3$$

$$= x^3 + 3hx^2 + 3h^2x + h^3$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$\frac{d}{dx}(x^3) = \lim_{h \rightarrow 0} \frac{(x^3 + 3hx^2 + 3h^2x + h^3) - x^3}{h}$$

$$= \lim_{h \rightarrow 0} \frac{3hx^2 + 3h^2x + h^3}{h}$$

$$= \lim_{h \rightarrow 0} (3x^2 + 3hx + h^2)$$

$$= 3x^2$$

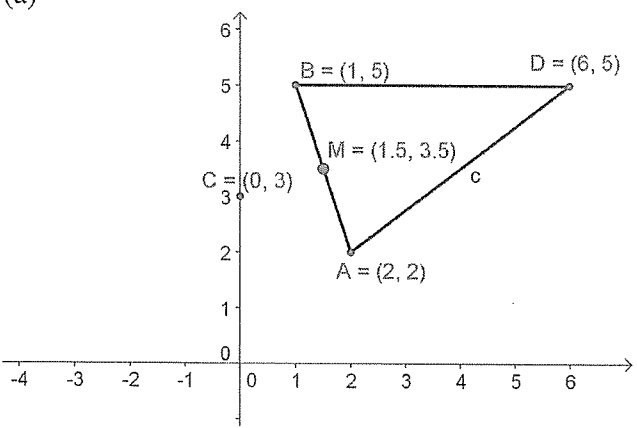
Too many students do not know definition of "from first principles" and use incorrect notation.

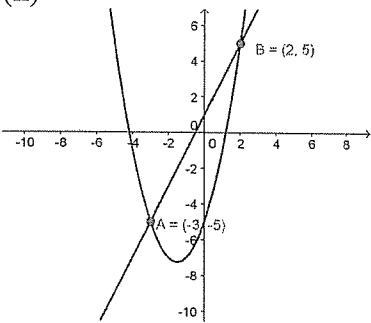
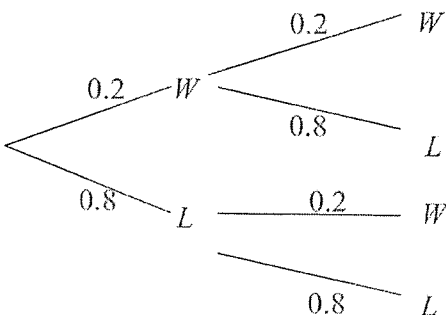
Question 12

Solutions	Marking Criteria												
(a) (i) $\frac{d}{dx}(e^x \sin x) = e^x \cos x + e^x \sin x$ $= e^x (\sin x + \cos x)$	① Correct product rule method ① Correct answer												
(ii) $\frac{d}{dx}\left(\frac{\log_e x}{x^3}\right) = \frac{x^3 \times \frac{1}{x} - 3x^2 \log_e x}{x^6}$ $= \frac{x^2 - 3x^2 \log_e x}{x^6}$ $= \frac{1 - 3 \log_e x}{x^4}$	① Correct rule ① Correct answer <i>Some students had numerator around the wrong way.</i>												
(b) $\int_0^{\frac{\pi}{4}} \frac{\cos x}{\sin x + 3} dx = \left[\ln(\sin x + 3) \right]_0^{\frac{\pi}{4}}$ $= \ln\left(\frac{1}{\sqrt{2}} + 3\right) - \ln 3$ $= \ln \frac{1 + 3\sqrt{2}}{\sqrt{2}} - \ln 3$ $= \ln\left(\frac{1 + 3\sqrt{2}}{3\sqrt{2}}\right)$ $= \ln\left(\frac{\sqrt{2} + 6}{6}\right)$	<i>A lot of students did not recognise</i> $\int \frac{f'(x)}{f(x)} dx = \ln(f(x)) + C$												
(c) (i) <table border="1"><tr><td>x</td><td>0</td><td>$\frac{\pi}{4}$</td><td>$\frac{\pi}{2}$</td><td>$\frac{3\pi}{4}$</td><td>π</td></tr><tr><td>y</td><td>1</td><td>$1\frac{1}{2}$</td><td>2</td><td>$1\frac{1}{2}$</td><td>1</td></tr></table> $A \approx \frac{h}{3}(y_0 + y_4 + 4 \times (y_1 + y_3) + 2 \times y_2)$ $= \frac{\pi}{3}\left(1 + 1 + 4 \times \left(1\frac{1}{2} + 1\frac{1}{2}\right) + 2 \times 2\right)$ (ii) $= \frac{\pi}{12}(2 + 12 + 4)$ $= \frac{\pi}{12} \times 18$ $= \frac{3\pi}{2}$ $\approx 4.71 \text{ units}^2$	x	0	$\frac{\pi}{4}$	$\frac{\pi}{2}$	$\frac{3\pi}{4}$	π	y	1	$1\frac{1}{2}$	2	$1\frac{1}{2}$	1	① All correct answers ① Correct formula & substitution ① Correct result
x	0	$\frac{\pi}{4}$	$\frac{\pi}{2}$	$\frac{3\pi}{4}$	π								
y	1	$1\frac{1}{2}$	2	$1\frac{1}{2}$	1								

(d) (i) Change solutions $T_1 = 3$ $T_2 = 5$ $T_3 = 7$ $a = 3, d = 2$ $T_n = 3 + 2(n-1)$ $= 3 + 2n - 2$ $= 2n + 1$ $2n + 1 = 45$ $2n = 44$ $n = 22$ $\therefore$ 22 regular hexagons	① Show pattern as an arithmetic sequence ① Correct answer <i>Some students did by trial/error OR counting which tended to lead to errors.</i>
(ii) $\text{Hexagon perimter} = 6 \times 3 + 6 \times 5 + \dots + 6 \times 45$ $= 6(3 + 5 + \dots + 45)$ $= 6 \times \frac{22}{2} (3 + 45)$ $= 66 \times 48$ $= 3168$ $\text{Total Length} = 3168 + 6 \times 45$ $= 3438 \text{ cm}$	① Show arithmetic series ① Correct formula ① Correct result <i>Some students forgot arms.</i> <i>Some failed to see question as an arithmetic series.</i> <i>Some had one sixth of answer for hexagons.</i>

Question 13

Solutions	Marking Criteria
(a)  (i) $M = \left(\frac{1+2}{2}, \frac{5+2}{2} \right)$ $= \left(\frac{3}{2}, \frac{7}{2} \right)$ (ii) $m_{AB} = \frac{5-2}{1-2}$ $= \frac{3}{-1}$ $= -3$ Required gradient is $\frac{1}{3}$ Equation of perpendicular bisector $y - \frac{7}{2} = \frac{1}{3} \left(x - \frac{3}{2} \right)$ $3y - \frac{21}{2} = x - \frac{3}{2}$ $6y - 21 = 2x - 3$ $0 = 2x - 6y + 18$ $x - 3y + 9 = 0$	① Required gradient ① Correct equation.
(iii) Let $x = 0$ $-3y + 9 = 0$ $y = 3$ $C = (0, 3)$	<i>Many students failed to recognise that every point on the perpendicular bisector of AB will be equidistant from A and B – so C is the y intercept of the perpendicular bisector.</i>
(iv) Sub $y = 5$ $x - 15 + 9 = 0$ $x = 6$ $D = (6, 5)$	

(v) $h = 5 - 2$ $= 3$ $b = 5$ $A = \frac{1}{2} \times 5 \times 3$ $= 7.5 \text{ units}^2$	① for any relevant and correct calculation. ① Correct answer <i>Much easier to do with the aid of a diagram.</i>
(b) (i) $y = x^2 + 3x - 5 \dots (1)$ $y = 2x + 1 \dots (2)$ Equating (1) and (2) $2x + 1 = x^2 + 3x - 5$ $0 = x^2 + x - 6$ $0 = (x + 3)(x - 2)$ $x = -3, x = 2$ When $x = 2, y = 5$ and $x = -3, y = -5$ The points of intersection are $(2, 5)$ and $(-3, -5)$	① for quadratic equation ① for points of intersection
(ii)  $A = \int_{-3}^2 ((2x + 1) - (x^2 + 3x - 5)) dx$ $= \int_{-3}^2 (-x^2 - x + 6) dx$ $= \left[-\frac{x^3}{3} - \frac{x^2}{2} + 6x \right]_{-3}^2$ $= \left(\frac{-8}{3} - 2 + 12 \right) - \left(\frac{27}{3} - \frac{9}{2} - 18 \right)$ $= \frac{125}{6} \text{ units}^2$	① for correct integration (including limits) ① for answer <i>Some students used the y values for the limits</i>
(c) 	① for either $P(WL)$ or $P(LW)$ ① Correct answer

$P(\text{one win}) = 0.2 \times 0.8 + 0.8 \times 0.2$ $= 2 \times 0.2 \times 0.8$ $= 0.32$	
(ii) Win first 0.2 Lost first win second 0.2×0.8 Lose first, lose second, win third $0.2 \times 0.8 \times 0.8$ $P(\text{at least one}) = 0.2 + 0.2 \times 0.8 + 0.2 \times 0.8^2 + \dots$ $a = 0.2, r = 0.8$ Let $S_n = \frac{9}{10}$ $\frac{9}{10} = \frac{0.2(1 - 0.8^n)}{1 - 0.8}$ $0.18 = 0.2 \times (1 - 0.8^n)$ $0.9 = 1 - 0.8^n$ $0.8^n = 0.1$ $n = \frac{\ln 0.1}{\ln 0.8}$ ≈ 10.31 Therefore need to play 11 matches to have a 90% chance of winning at least one game. <u>ALTERNATIVE METHOD:</u> $P(\text{win at least one}) = 1 - P(\text{lose all})$ $0.9 = 1 - (0.8)^n$ $0.8^n = 0.1$ $\ln 0.8^n = \ln 0.1$ $n \ln 0.8 = \ln 0.1$ $n = \frac{\ln 0.1}{\ln 0.8}$	① for setting up $0.8^n = 0.1$ ① Correct answer <i>Over-simplified by many students and poorly done.</i>

Question 14

Solutions	Marking Criteria
(a) (i) $y = x^3 + 4x^2 - 3x + 2$ $\frac{dy}{dx} = 3x^2 + 8x - 3$ $\frac{d^2y}{dx^2} = 6x + 8$	① for $\frac{dy}{dx}$ ① for $\frac{d^2y}{dx^2}$
(ii) Let $\frac{dy}{dx} = 0$ $3x^2 + 8x - 3 = 0$ $3x^2 + 9x - x - 3 = 0$ $3x(x+3) - 1(x+3) = 0$ $(3x-1)(x+3) = 0$ $x = -3, x = \frac{1}{3}$ When $x = -3$ $y = (-3)^3 + 4(-3)^2 - 3(-3) + 2$ $= 20$ When $x = \frac{1}{3}$ $y = \left(\frac{-1}{3}\right)^3 + 4\left(\frac{-1}{3}\right)^2 - 3\left(\frac{-1}{3}\right) + 2$ $= \frac{40}{27} (\approx 1.4)$ Testing: When $x = -3$ $\frac{d^2y}{dx^2} = 6(-3) + 8$ $= -10 < 0$ Therefore maximum at $(-3, 20)$ When $x = \frac{1}{3}$ $\frac{d^2y}{dx^2} = 6\left(\frac{1}{3}\right) + 8$ $= 10 > 0$ Therefore minimum at $\left(\frac{1}{3}, \frac{40}{27}\right)$	① for minimum stat. pt ① for maximum stat. pt <i>Students are reminded that if they are using a table of values to determine the nature of the stationary point, NOT to just use + and - signs in the table.</i>

$\therefore$ Initial mass is 20 kg

(ii) $M = 20e^{-kt}$ When $t = 200$, $M = 10$ $10 = 20e^{-200k}$ $\frac{1}{2} = e^{-200k}$ $-200k = \ln \frac{1}{2}$ $k = \frac{-1}{200} \ln \frac{1}{2}$ $k = \frac{1}{200} \ln 2$	① for setting up $\frac{1}{2} = e^{-200k}$ ① Correct answer in exact form
(iii) Let $t = 2000$ $M = 20 \times e^{-2000 \times \frac{1}{200} \ln 2}$ $= 0.1953...$ $= 19.53... \text{ grams}$ $= 20 \text{ grams (nearest gram)}$	
(c) $x^2 - 2x + 3 - k = 0$ $\Delta = 4 - 4 \times 1 \times (3 - k)$ Let $\Delta \geq 0$ $0 \leq 4 - 4(3 - k)$ $0 \leq 4 - 12 + 4k$ $4k \geq 8$ $k \geq 2$	① for discriminant OR stating condition for real roots to exist. <i>Many students failed to find the discriminant correctly.</i> <i>Students who incorrectly stated that for real roots, $\Delta > 0$ lost a mark.</i>

Question 15

Solutions	Marking Criteria
(a) (i) In $\triangle ADF$ and $\triangle BCE$ $\leftarrow$ 1. $AD = BC$ (sides of a square equal) } ① 2. $\angle ADF = 90^\circ = \angle BCE$ (property of a square) } ① $DF = DE + EF$ $CE = CF + EF$ 3. But $DE = CF$ (given) } ① (with reasonable explanation) $\therefore DF = CF + EF = CE$ $DF = CE$ $\therefore \triangle ADF \equiv \triangle BCE$ (SAS) ①	Some students did not write introduction. Some students wrote RHS when there was <u>no</u> hypotenuse.
(ii) $\angle AFD = \angle BEF$ (corresponding angles, $\triangle ADF \equiv \triangle BCE$) Let $\angle AFD = \beta = \angle BEF$ } ① $\angle GFE = \beta = \angle GEF$ (same angles) } $\therefore FE = GF$ (equal sides opposite equal angles in an isosceles triangle)	Some students tried to use $AF = BE$ $\therefore GE = EF$ (not acceptable). This assumes $AG = BG$ which was not proved.
(b) (i) Let $\frac{dx}{dt} = 0$ $t^2 - 5t + 6 = 0$ $(t - 3)(t - 2) = 0$ $t = 2, t = 3$ Velocity is zero at 2 seconds and 3 seconds.	Too many incorrectly factored $t^2 - 5t + 6$ to $(t - 6)(t + 1)$
(ii) $\frac{d^2x}{dt^2} = 2t - 5$ Let $\frac{d^2x}{dt^2} = 0$ $t = \frac{5}{2}$ When $t = \frac{5}{2}$ $\frac{dx}{dt} = \left(\frac{5}{2}\right)^2 - 5\left(\frac{5}{2}\right) + 5$ $= -0.25$ Minimum velocity is -0.25 metres per second	Too many did not realise that $\frac{dx}{dt}$ was parabolic $\therefore$ min was at $t = 2\frac{1}{2}$ Too many used $t = 2$ and $t = 3$ as times when velocity was a min. from previous question.
(iii) $x = \frac{t^3}{3} - \frac{5t^2}{2} + 6t + C$ When $t = 0, x = -3$ $-3 = C$ $x = \frac{t^3}{3} - \frac{5t^2}{2} + 6t - 3$	A few students used $t = 0, x = 3$ (misread question) ① for constant ① for correct expression for x
(iv)	Since particle changed directions at

When $t = 0, x = -3$ When $t = 2$ $x = \frac{2^3}{3} - \frac{5 \times 2^2}{2} + 6 \times 2 - 3$ $= \frac{5}{3}$ When $t = 3$ $x = \frac{3^3}{3} - \frac{5 \times 3^2}{2} + 6 \times 3 - 3$ $= 1.5$ Distance travelled is $3 + \frac{5}{3} + \frac{5}{3} - 1.5$ $= \frac{29}{6}$ Distance travelled is $\frac{29}{6}$ metres	$t = 2$ and $t = 3$ and started at $t = 0$, we need positions for each of these times to see where movement was made. <u>ALTERNATIVELY:</u> $\text{Distance} = \left \int_0^2 f(t) dt \right + \left \int_2^3 f(t) dt \right $ $= \frac{29}{6} \text{ m}$
(c) (i) $A_1 = 30000 \times \left(1 + \frac{0.09}{12}\right) - Q$ $= 30000 \times 1.0075 - Q$	<i>Some confused this as as a savings question (i.e. superannuation)</i>
(ii) $A_2 = 1.0075 \times A_1 - Q$ $= 1.0075 \times (1.0075 \times 30000 - Q) - Q$ $= 30000 \times 1.0075^2 - Q(1 + 1.0075)$ $A_3 = 1.0075 \times A_2$ $= 1.0075(30000 \times 1.0075^2 - Q(1 + 1.0075)) - Q$ $= 30000 \times 1.0075^3 - Q(1 + 1.0075 + 1.0075^2)$ $A_n = 1.0075^n \times 30000 - Q(1 + 1.0075 + 1.0075^2 + \dots + 1.0075^{n-1})$ $A_n = 1.0075^n \times 30000 - \frac{Q \times (1.0075^n - 1)}{1.0075 - 1}$ $= 1.0075^n \times 30000 - Q \times \frac{400}{3} (1.0075^n - 1)$ $= 30000(1.0075)^n - \frac{400Q(1.0075^n - 1)}{3}$	<i>To show a pattern, A_1, A_2 & A_3 should be shown (A_1 above)</i> ① for A_n in this form <i>Preferable if explanation was inserted here i.e. a geometric series with $a = 1$, $r = 1.0075$, $n = n$ (although no deduction was made for not doing so)</i> ① for correct expression
(iii) When $n = 60$, $A_{60} = 0$	① Correct equation ① Correct answer <i>Mostly well done question.</i>

$$0 = 30000(1.0075)^{60} - \frac{400Q(1.0075^{60} - 1)}{3}$$

$$\frac{400Q(1.0075^{60} - 1)}{3} = 30000(1.0075)^{60}$$

$$Q = \frac{3 \times 30000(1.0075)^{60}}{400(1.0075^{60} - 1)}$$

$$= \$622.75 \text{ (nearest cent)}$$

Question 16

Solutions	Marking Criteria
(a) $6x^2 - 11 \equiv A(x+2)^2 + Bx + C$ $= A(x^2 + 4x + 4) + Bx + C$ $= Ax^2 + 4Ax + 4A + Bx + C$ $= Ax^2 + (4A + B)x + (4A + C)$ Equating coefficients: $A = 6 \qquad 4A + B = 0 \qquad 4A + C = -11$ $4(6) + B = 0 \qquad 4(6) + C = -11$ $B = -24 \qquad C = -35$	<i>Quadratic identity very poorly answered.</i>
(b) (i) $a^2 + b^2 = 7ab$ $(a+b)^2 = a^2 + 2ab + b^2$ $= 7ab + 2ab$ $= 9ab$ $\frac{(a+b)^2}{9} = ab$ $\left(\frac{a+b}{3}\right)^2 = ab$	<i>Not well answered – some students find difficult to SHOW things.</i>
(ii) $\log\left(\frac{a+b}{3}\right) - \frac{1}{4}(\log a + \log b) = \log\sqrt{ab} - \frac{1}{4}\log ab$ $= \frac{1}{2}\log ab - \frac{1}{4}\log ab$ $= \frac{1}{4}\log ab$	
(c) (i) Let $\angle ABE = \alpha$ $AD^2 = 25$ $AE^2 + DE^2 = 16 + 9$ $= 25$ $\therefore \angle AED = 90^\circ$ ($\triangle AED$ right-angled triangle) Let $\angle FBE = \alpha$ $\angle BAF + \angle AFB + \angle FBA = 180^\circ \text{ (}\angle \text{ sum of triangle)}$ $\angle BAF + 90^\circ + \alpha = 180^\circ$ $\angle BAF = 90^\circ - \alpha$ $\angle DEF = \angle BAE \text{ (alternate } \angle \text{'s, } DC \parallel AB)$ $\therefore \angle BAE = 90^\circ$	<i>Some terribly <u>long</u> solutions.</i> <i>Students must give <u>logical reasons</u> and try to be <u>concise</u>.</i>

$\angle BAE = \angle EAF + \angle FAB$ $90^\circ = \angle EAB + 90^\circ - \alpha$ $\angle EAB = \alpha$	
(ii) In $\triangle AED$ $\cos \angle ADE = \frac{4}{5}$ $\angle ADE = \cos^{-1}\left(\frac{4}{5}\right)$ $EF^2 = 9 + 25 - 2 \times 3 \times 5 \cos \left[\cos^{-1}\left(\frac{4}{5}\right) \right]$ $= 9 + 25 - 2 \times 3 \times 4$ $= 10$ $EF = \sqrt{10}$	Most did not realise to use the cosine rule – some tried Pythagoras (but no right angle!)
(d) (i) $AB + AC + \text{arc}BC = 12$ $r + r + r\theta = 12$ $2r + r\theta = 12$ $r(2 + \theta) = 12$ $r = \frac{12}{2 + \theta}$ $Y = \frac{1}{2}r^2\theta$ $Y = \frac{1}{2} \times \left(\frac{12}{2 + \theta} \right)^2 \times \theta$ $= \frac{144}{2(2 + \theta)^2} \times \theta$ $= \frac{72\theta}{(2 + \theta)^2}$	
(ii) $Y = \frac{72\theta}{(2 + \theta)^2}$ $\frac{dY}{d\theta} = \frac{(2 + \theta)^2 \times 72 - 72\theta \times 2(2 + \theta)}{(2 + \theta)^4}$ Let $\frac{dY}{d\theta} = 0$ $\frac{(2 + \theta)^2 \times 72 - 72\theta \times 2(2 + \theta)}{(2 + \theta)^4} = 0$ $72(\theta + 2)[\theta + 2 - 2\theta] = 0$ $72(\theta + 2)(2 - \theta) = 0$ $\theta = 2 (\theta > 0)$	

Checking if maximum

θ	1	2	3
$\frac{dY}{d\theta}$	$\frac{8}{3}$	0	$\frac{-72}{125}$

Therefore maximum with $\theta = 2$

$$\begin{aligned} Y &= \frac{72 \times 2}{(2+2)^2} \\ &= \frac{144}{16} \\ &= 9 \text{ m}^2 \end{aligned}$$

Many students did not attempt to show max, whilst many did not show values in their table.